

Middle Convolution of ℓ -adic Perverse Sheaves

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1 Introduction

1.1 Notations

1.1.1. Motivated by the growing interest in the formalization of mathematics within the mathematical community, we adopt a style of notation inspired by type theory throughout this write-up, without committing to type-theoretic foundations. The symbols below denote types and/or categories, depending on your philosophical inclination. The notation $X : T$ means that X is a term of type T or an object of the category T .

Set	Sets
Set^{fin}	Finite sets
Top	Topological spaces
$\text{Open}(T)$	Open subsets of $T : \text{Top}$
Grp	Groups
Grp^{top}	Topological groups
Grp^{pro}	Profinite groups
Grp^{ab}	Abelian groups
Ring	Rings (associative, unital, not necessarily commutative)
Ring^{top}	Topological rings
Field	Fields
$\text{Field}^{\text{top}}$	Topological fields
Mod_R	Left modules over $R : \text{Ring}$
Vect_k	Vector spaces over $k : \text{Field}$
$\text{Vect}_k^{\text{fin}}$	Finite-dimensional vector spaces over $k : \text{Field}$
$\text{Rep}_\Lambda(G)$	Continuous Λ -linear representations of $G : \text{Grp}^{\text{top}}$, where $\Lambda : \text{Ring}^{\text{top}}$
$\text{Rep}_\Lambda^{\text{fin}}(G)$	Continuous Λ -linear representations of $G : \text{Grp}^{\text{top}}$, where $\Lambda : \text{Ring}^{\text{top}}$, whose underlying topological Λ -modules are finitely generated
Sch	Schemes
Sch_S	Schemes over a base scheme $S : \text{Sch}$
Var_k	Varieties over $k : \text{Field}$, i.e. integral separated schemes of finite type over k
Curv_k	Curves over $k : \text{Field}$, i.e. one-dimensional varieties over k
$\text{Curv}_k^{\text{sm}}$	Smooth curves over $k : \text{Field}$
Riem	Riemann surfaces, i.e. connected one-dimensional complex manifolds
Site	Sites, i.e. categories equipped with Grothendieck topologies
$\text{Sh}(X, A)$	Sheaves on $X : \text{Site}$ with values in a category A admitting small limits
$\text{Mod}(X, \mathcal{R})$	Sheaves of left $\mathcal{R}$ -modules on $X : \text{Site}$, where $\mathcal{R} : \text{Sh}(X, \text{Ring})$
$D^b(A)$	Bounded derived objects of an abelian category A
$D^b(X, k)$	Bounded derived objects of $\text{Sh}(X, \text{Vect}_k)$, where $X : \text{Site}$ and $k : \text{Field}$
$X \rightarrow Y$	Morphisms from $X : T$ to $Y : T$

1.2 Acknowledgements

1.2.1. The author thanks his advisor, Prof. Chris Hall, for his guidance and advice.

1.2.2 Typesetting. The typesetting style of this write-up is inspired by [Kat88].

1.2.3 Declaration of AI Use. The use of generative AI in the preparation of this write-up was limited to the following purposes: polishing wording, phrasing, grammar, and formatting; offering writing advices; answering mathematical questions from the author to help him learn the subject; searching for and citing appropriate references; translating reading materials in French into English; retypesetting old references with archaic typesetting for ease of reading; providing emotional support for the author; providing fitness advice to help the author stay healthy and energetic while working on the write-up.

1.3 Motivations

1.3.1. *Monodromy groups* are interesting mathematical objects that arise naturally in both geometric and arithmetic contexts. Roughly speaking, they describe how locally defined data change when transported along a loop. Controlling and understanding monodromy groups has been a central theme in both of these worlds, with significant applications.

1.3.2. The goal of this write-up is to provide an exposition on monodromy groups as well as ways to study them using methods from perverse sheaves and their middle convolution introduced by Katz [Kat96], especially in the ℓ -adic arithmetic context.

1.3.3. Consider the equation

$$z(w)^2 = w$$

for an unknown holomorphic function $z : V \rightarrow \mathbb{C}^\times$, where $V : \text{Open}(\mathbb{C}^\times)$ and $w : V$ is the independent variable. Fix a base point $w_0 : \mathbb{C}^\times$ and choose $z_0 : \mathbb{C}^\times$ with $z_0^2 = w_0$. On a sufficiently small disc V containing w_0 , the equation has two holomorphic solution branches z_+ and z_- , uniquely specified by $z_+(w_0) = z_0$ and $z_-(w_0) = -z_0$. Move w once around 0 along the loop

$$\gamma(t) = w_0 e^{2\pi i t} \quad t : [0, 1].$$

The continued values of the two branches along this loop are $\tilde{\gamma}_+(t) = z_0 e^{\pi i t}$ and $\tilde{\gamma}_-(t) = -z_0 e^{\pi i t}$. At $t = 1$, they have exchanged their initial values, so analytic continuation exchanges the solution germs $[z_+]_{w_0}$ and $[z_-]_{w_0}$. In other words, there is a permutation representation of the fundamental group¹ $\pi_1^{\text{top}}(\mathbb{C}^\times, w_0) \cong \mathbb{Z} : \text{Grp}^{\text{top}}$

$$\rho : \pi_1^{\text{top}}(\mathbb{C}^\times, w_0) \rightarrow \text{Aut}_{\text{Set}}(\{[z_+]_{w_0}, [z_-]_{w_0}\}) \quad n \mapsto ([z_+]_{w_0} [z_-]_{w_0})^n.$$

We call this representation the *permutation monodromy representation* and its image, S_2 , the *permutation monodromy group* of the equation.

1.3.4. Let us reformulate the above in terms of sheaves. Consider the covering

$$f : \mathbb{C}^\times \rightarrow \mathbb{C}^\times \quad \zeta \mapsto \zeta^2.$$

¹with the discrete topology

Let $\mathcal{F} : \text{Sh}(\mathbb{C}^\times, \text{Set})$ be its sheaf of holomorphic sections. Equivalently, this is the solution sheaf of $z(w)^2 = w$: for $V : \text{Open}(\mathbb{C}^\times)$,

$$\begin{aligned}\mathcal{F}(V) &= \{z : V \rightarrow \mathbb{C}^\times \mid z \text{ holomorphic and } f \circ z = 1_V\} \\ &= \{z : V \rightarrow \mathbb{C}^\times \mid z \text{ holomorphic and } z(w)^2 = w \text{ for every } w : V\}.\end{aligned}$$

Here $f \circ z = 1_V$ means that the composite takes values in V and, regarded as a map $V \rightarrow V$, is the identity. On a sufficiently small disc V containing w_0 , the two branches z_+, z_- give an isomorphism $\mathcal{F}|_V \cong \{+, -\}_V$, so $\mathcal{F}$ is locally constant. Its stalk is $\mathcal{F}_{w_0} = \{[z_+]_{w_0}, [z_-]_{w_0}\} \cong f^{-1}(w_0)$, with the last identification given by evaluation. Let $\Pi_1(X)$ denote the fundamental groupoid of a topological space $X : \text{Top}$, whose objects are points of X and whose morphisms are homotopy classes of paths relative to their endpoints, with concatenation as composition. Analytic continuation gives a *transport functor*

$$T : \Pi_1(\mathbb{C}^\times) \rightarrow \text{Set} \quad w \mapsto \mathcal{F}_w \quad [\gamma] \mapsto T_\gamma,$$

where, for a path $\gamma : [0, 1] \rightarrow \mathbb{C}^\times$, the *transport map* is

$$T_\gamma : \mathcal{F}_{\gamma(0)} \rightarrow \mathcal{F}_{\gamma(1)} \quad [z]_{\gamma(0)} \mapsto [z^\gamma]_{\gamma(1)}.$$

Here $[z^\gamma]_{\gamma(1)}$ is the solution germ obtained by continuing $[z]_{\gamma(0)}$ along γ . Under evaluation, this is precisely path lifting through f : the lift starting at $z(\gamma(0))$ ends at $z^\gamma(\gamma(1))$. Restricting transport to loops at w_0 gives the permutation monodromy representation

$$\rho : \pi_1^{\text{top}}(\mathbb{C}^\times, w_0) \rightarrow \text{Aut}_{\text{Set}}(\mathcal{F}_{w_0}) \quad [\gamma] \mapsto T_\gamma.$$

1.3.5. The sheaf $\mathcal{F}$ is locally constant but not constant: a global section would determine a germ at w_0 fixed by all monodromy, whereas transport once around 0 exchanges the two square-root germs, as demonstrated in 1.3.3. Thus $\mathcal{F}(\mathbb{C}^\times) = \emptyset$. Locally constant (but not necessarily constant) sheaves of sets, such as this one, are known as *local systems (of sets)*.

1.3.6. We have looked at permutation monodromy, but permutation actions do not capture all the situations we wish to study, since a lot of times in nature we encounter actions on vector spaces instead of sets. For example, instead of an algebraic equation like the one in 1.3.3, we consider a differential equation, which is known as *Gauss's hypergeometric equation*

$$\lambda(1 - \lambda) \frac{d^2 f}{d\lambda^2} + (c - (a + b + 1)\lambda) \frac{df}{d\lambda} - abf = 0 \quad (1.3.6.1)$$

where $a, b, c : \mathbb{C}$ are fixed parameters and $f : V \rightarrow \mathbb{C}$ is an unknown holomorphic function, with $V : \text{Open}(U)$, $U = \mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\}$, and independent variable $\lambda : V$. We then define the sheaf of solutions to this equation $\mathcal{L} : \text{Sh}(U, \text{Vect}_{\mathbb{C}})$

$$\mathcal{L}(V) = \{f : V \rightarrow \mathbb{C} \mid f \text{ holomorphic and satisfies (1.3.6.1) on } V\}.$$

This is a sheaf of vector spaces since (1.3.6.1) is a homogeneous linear differential equation. For each $\lambda_0 : U$, the existence-and-uniqueness theorem [HTT08, Theorem 4.3.1] gives a sufficiently small open disc $V : \text{Open}(U)$ containing λ_0 and an isomorphism of sheaves

$$\underline{\mathbb{C}}_V^2 \xrightarrow{\sim} \mathcal{L}|_V$$

Thus $\mathcal{L}$ is locally constant of rank 2. We call such a sheaf a *complex local system of rank 2*.

1.3.7. As in 1.3.4, analytic continuation gives a *transport functor*

$$T : \Pi_1(U) \rightarrow \text{Vect}_{\mathbb{C}}^{\text{fin}} \quad \lambda \mapsto \mathcal{L}_{\lambda} \quad [\gamma] \mapsto T_{\gamma},$$

where, for a path $\gamma : [0, 1] \rightarrow U$,

$$T_{\gamma} : \mathcal{L}_{\gamma(0)} \rightarrow \mathcal{L}_{\gamma(1)} \quad [f]_{\gamma(0)} \mapsto [f^{\gamma}]_{\gamma(1)}.$$

Here $[f^{\gamma}]_{\gamma(1)}$ is the solution germ obtained by continuing $[f]_{\gamma(0)}$ along γ . Since continuation preserves linear combinations, each T_{γ} is a linear isomorphism. Restricting transport to loops at a base point $\lambda_0 : U$ gives the *linear monodromy representation*

$$\rho : \pi_1^{\text{top}}(U, \lambda_0) \rightarrow \text{Aut}_{\text{Vect}_{\mathbb{C}}}(\mathcal{L}_{\lambda_0}) \quad [\gamma] \mapsto T_{\gamma}.$$

1.3.8 A Question in Arithmetic Geometry. Here is a

2 Local Systems and Lisse Sheaves

2.1 Local Systems in Topology

2.1.1 Notation. We use the following notation:

$X : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	A smooth projective connected curve over $\mathbb{C}$
$S : \text{Set}^{\text{fin}}$	A finite nonempty subset of closed points of X
$U : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	The open subvariety $X \setminus S$
$j : U \rightarrow X$	The inclusion map
$m : \mathbb{N}$	The cardinality of S
$(-)^{\text{an}} : \text{Curv}_{\mathbb{C}}^{\text{sm}} \rightarrow \text{Riem}$	The complex analytification functor

2.2 Local Systems in Complex Algebraic Geometry

2.3 Local Systems in ℓ -adic Arithmetic Geometry: Lisse Sheaves

2.4 Rigidity of Local Systems

3 Perverse Sheaves and Middle Convolution

3.1 Six-Functor Formalism

3.2 Middle Extension Sheaves

3.3 Middle Convolution

4 Mod- ℓ Monodromy of Hyperelliptic Curves

References

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